someRpackages

Simon Labarthe

August 2026

Contents

1 Abstract **1**

2 Introduction **2**

3 Package Overview and Core Functions **3**

3.1 Installation and Quick Start 3

3.2 Available Functions 3

3.3 Design Philosophy 3

4 A Worked Example: Starwars **4**

4.1 Descriptive Table 4

4.2 Frequency Table 5

4.3 AI-Powered Text Categorisation 5

5 Closing Statement **7**

1 Abstract

This article introduces `someRpackages`, a curated collection of R functions designed to enhance efficiency in data analysis, automated reporting, and AI-supported text classification. Researchers frequently face repetitive tasks in data preparation and output generation. The package addresses this by offering straightforward functions for creating publication-ready LaTeX/PDF tables and integrating with AI APIs for content coding. The core functions are presented with practical examples obviously coming from `starwars`. By automating these common analytical workflows, `someRpackages` allows researchers to focus on substantive interpretation rather than technical implementation. The package is open-source and available on GitHub.

Keywords. R package, Data Analysis, Automated Reporting, LaTeX, Text Classification, API.

2 Introduction

Social scientists spend a substantial portion of their time not on thinking, but on the mechanical aspects of data analysis: cleaning data, writing repetitive code to generate tables, and manually classifying text data. This is an inefficient use of intellectual resources (already limited in my case). While R offers powerful analytical capabilities, the path from a clean dataset to a polished, publication-ready table is often fraught with cumbersome code and debugging.

`someRpackages` offers a pragmatic solution by wrapping complex R operations into intuitive functions that produce LaTeX and PDF outputs.

3 Package Overview and Core Functions

`someRpackages` is a collection of R utility functions designed for data analysis and automated reporting. The package is designed to be intuitive, requiring minimal R expertise to generate high-quality outputs.

3.1 Installation and Quick Start

The package can be loaded directly from GitHub using the provided script. To get started, run the following in your R console:

```
# Load and install all functions with one line
source("https://raw.githubusercontent.com/SimonLabarthe/someRpackages/main/load_someRpackages.R")
load_someRpackages()
```

This command sources and loads the core functions into the global environment. For PDF and LaTeX output, a LaTeX distribution is required, which can be conveniently installed via R:

```
install.packages("tinytex")
tinytex::install_tinytex()
```

3.2 Available Functions

As of September 10, 2026, the package includes the following key functions:

1. `create_descriptive_table()`: Generates a comprehensive descriptive statistics table (including n, mean, median, mode, SD, min, max, and quartiles) for a given dataset. It creates a LaTeX/.tex file and compiles it into a standalone PDF, offering extensive customization options (e.g., selecting which statistics to display).
2. `create_frequency_table_2vars()`: Creates a cross-tabulation frequency table for two variables, outputting results in a similar polished PDF format.
3. `create_frequency_table_3vars()`: Extends the frequency table functionality to three variables, producing a nested frequency table that is particularly useful for subgroup analysis.
4. `categorise_api()`: An advanced function that uses the DeepSeek API to classify a vector of texts. It allows researchers to efficiently code large text corpora based on a custom taxonomy, with built-in checkpointing and batch processing.

3.3 Design Philosophy

The design philosophy behind `someRpackages` is guided by two key principles. First, **laziness**: each function is designed to perform a complete task with minimal user input. The goal is to reduce complex data processing and output generation to a single readable line of code. For example, generating a full descriptive statistics table in PDF is as simple as:

```
create_descriptive_table(data, output_dir = getwd())
```

Second, **maximum coolness**: the emphasis on LaTeX and PDF generation ensures that the output is ready for immediate integration into manuscripts, reports, or presentations. This approach bridges the gap between data analysis and scholarly publication.

4 A Worked Example: Starwars

To illustrate the package in action, we use the `starwars` dataset bundled with `dplyr`.

```
library(dplyr)
## Warning: package 'dplyr' was built under R version 4.5.3
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##     filter, lag
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
data("starwars", package = "dplyr")
```

4.1 Descriptive Table

```
create_descriptive_table(
  starwars,
  output_name = "starwars_descriptives",
  output_dir  = getwd(),
  mean        = TRUE,
  median      = TRUE,
  sd          = TRUE,
  min         = TRUE,
  max         = TRUE,
  mode        = TRUE
)
```

```
starwars %>%
  select(height, mass, birth_year) %>%
  summary()
##      height           mass           birth_year
##  Min.   : 66.0   Min.   :  15.00   Min.   :  8.00
```

```
##  1st Qu.:167.0   1st Qu.:  55.60   1st Qu.: 35.00
##  Median :180.0   Median :  79.00   Median : 52.00
##  Mean   :174.6   Mean   :  97.31   Mean   : 87.57
##  3rd Qu.:191.0   3rd Qu.:  84.50   3rd Qu.: 72.00
##  Max.   :264.0   Max.   :1358.00   Max.   :896.00
##  NA's   :6       NA's   :28        NA's   :44
```

4.2 Frequency Table

```
create_frequency_table_2vars(starwars, var1 = "gender", var2 = "species")
```

```
starwars %>%
  count(gender, species) %>%
  head(10)
```

gender	species	n
feminine	Clawdite	1
feminine	Droid	1
feminine	Human	9
feminine	Kaminoan	1
feminine	Mirialan	2
feminine	Tholothian	1
feminine	Togruta	1
feminine	Twi'lek	1
masculine	Aleena	1
masculine	Besalisk	1

4.3 AI-Powered Text Categorisation

The `categorise_api()` function classifies a vector of texts via the DeepSeek API.

```
my_texts <- c(
  "The new childcare policy will reduce costs for families.",
  "We must fight for equal pay for equal work.",
  "The traditional family unit is the foundation of our society."
)

api_key <- Sys.getenv("DEEPSEEK_API_KEY")

results <- categorise_api(
```

```
  texts           = my_texts,
  api_key         = api_key,
  batch_size      = 2,
  checkpoint_file = "my_categorisation.rds"
)

print(results)
```

The function returns a data frame with the same number of rows as the input texts, containing the predicted `category` and `theme` for each text.

5 Closing Statement

I believe that `someRpackages` can be a valuable asset for the research community. I invite researchers to use it, contribute to it, and adapt it to their needs. The source code is freely available on GitHub, and I welcome any feedback or contributions :)

5.0.1 Citation

It took me a while to create all of this. I would really appreciate it if you could make my work more visible.

You are pleased to cite as follow:

Labarthe, S. (2026). someRpackages: R Utility Functions. https://github.com/SimonLabarthe/someRpackages

Visit My Website https://github.com/SimonLabarthe